

SOEN 6011 D2: F6 Beta Function

From-Scratch Numerical Core, Tkinter GUI, Version Control, and Updated Requirements

Wei Huang

Concordia University

Summer 2026

D2 Scope and Dependency Chain

D1 baseline

- Log-gamma identity selected
- Python TUI
- `math.lgamma` and `math.exp`

D2 increment

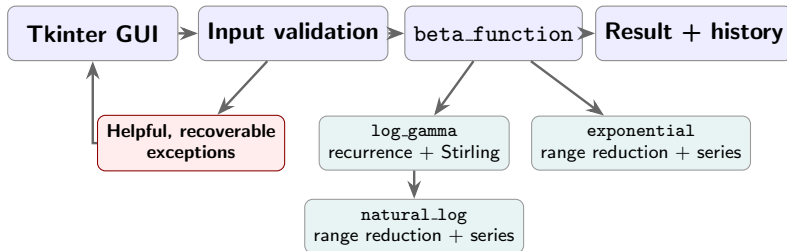
- 1 **P5:** implement mathematics from scratch
- 2 **P5:** replace TUI with Tkinter GUI
- 3 **P6:** publish Git repository + README
- 4 **P7:** revise requirements

Interpretation of “from scratch”

No `math` module or numerical library. Only input/output, arithmetic, exception handling, and Tkinter UI services are used. (Kamthan, 2026a)

D2 revisits the D1 solution and adds new behavior, following an iterative and incremental process. (Kamthan, 2026b)

P5: From-Scratch Architecture



Subordinate functions implemented from scratch: `natural_log`, `exponential`, and `log_gamma`.

Allowed dependencies

`tkinter`, `ttk`, float parsing, formatting, arithmetic.

Prohibited dependencies avoided

No `gamma`, `lgamma`, `log`, `exp`, `SciPy`, or `NumPy`.

P5: Numerical Method and Verification

Subordinate mathematics

$$\ln m = 2 \sum_{k=0}^{\infty} \frac{z^{2k+1}}{2k+1}, \quad z = \frac{m-1}{m+1}, \quad 1 \leq m < 2$$

$$e^r = \sum_{k=0}^{\infty} \frac{r^k}{k!}$$

Shift x to $x \geq 8$, then use the Gamma recurrence and a Stirling correction series.

$$\log B = \log \Gamma(x) + \log \Gamma(y) - \log \Gamma(x+y)$$

All verification values were produced externally and used only for validation. The submitted implementation itself imports no mathematical library. (NIST DLMF, 2026a; NIST DLMF, 2026b)

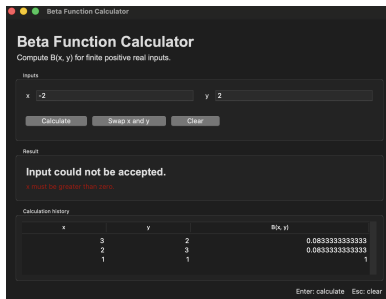
External verification only

Input	Computed value	Rel. error
$B(1, 1)$	0.999999999999988	1.24×10^{-14}
$B(2, 3)$	0.0833333333333322	1.45×10^{-14}
$B(0.5, 0.5)$	3.14159265358979	1.70×10^{-15}
$B(100, 300)$	$5.94745626861 \times 10^{-99}$	9.40×10^{-13}

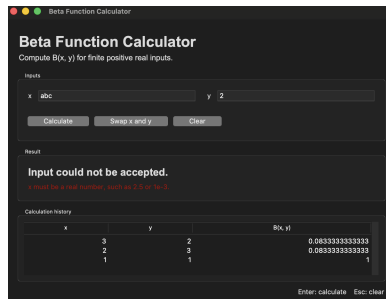
Numeric limit

A representable finite result is returned; otherwise a `NumericRangeError` explains overflow or underflow.

P5: Tkinter GUI and Exception Recovery



Domain error: names x and states the $x > 0$ correction.

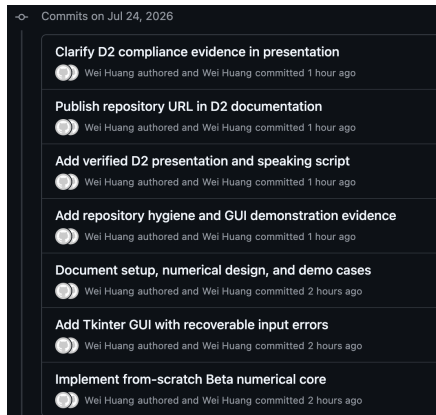


Nonnumeric error: names x and provides valid examples.

Implementation and recovery evidence

GUI implemented entirely with Tkinter widgets. Both screenshots show that handled errors preserve calculation history and leave the window ready for corrected input.

P6: Distributed Version Control Evidence



High-quality history

Purpose-focused commits separate:

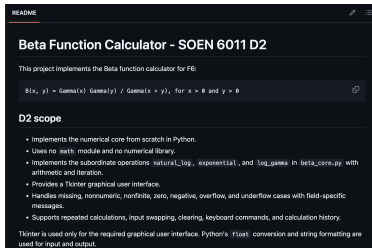
- from-scratch numerical core
- Tkinter GUI and recovery
- README and demonstration evidence
- presentation and repository publication

Public repository

`https://github.com/HW925/SOEN-6011-D2-Beta-Function`

Commit logs make implementation decisions and increments explicit. (Kamthan, 2026c)

P6: README Documentation Evidence



Scope: from-scratch boundary, subordinate functions, Tkinter, and handled cases.

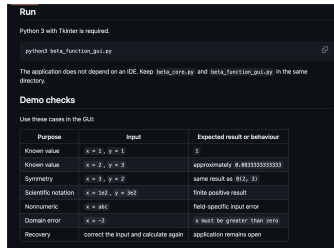
Numerical design

`natural_log` reduces its argument to `[1, 2)` and evaluates an `atanh`-based series. `exponential` reduces its argument using powers of two and evaluates a Taylor series. `log_gamma` shifts positive inputs to at least 8 using the Gamma recurrence and then applies a Stirling series. The Beta value is evaluated in the logarithmic domain to avoid directly multiplying large Gamma values.

Floating-point range still applies. The program reports a numeric-range error when the final result cannot be represented as a nonzero finite float.

Repository

Public repository URL: <https://github.com/HW925/SOEN-6011-D2-Beta-Function>



Usage: IDE-independent run command and reproducible demo checks.

README coverage

Numerical method, floating-point limit, repository URL, author, and execution instructions are documented alongside the source.

P7: Updated Functional Requirements

Legend: FR = Functional Requirement. These replace the D1 TUI-focused functional requirements.

GUI interaction and calculation

D2-FR-01 The system shall provide labeled GUI fields for x and y , plus Calculate, Swap, and Clear commands.

D2-FR-02 The system shall accept finite positive real inputs in decimal or scientific notation.

D2-FR-03 The system shall compute $B(x, y)$ through the from-scratch `natural_log`, `exponential`, and `log_gamma` operations.

D2-FR-04 The system shall display each representable result with at least ten significant digits.

D2-FR-05 The system shall preserve a history of successful calculations during the current session.

D2-FR-06 The system shall allow another calculation after a result or handled error without restarting.

Why requirements changed

D1's terminal prompts and quit command were replaced by GUI controls, persistent history, and recovery behavior required by P5.

P7: Updated Constraints and Quality Requirements

Constraints

D2-CR-01 The GUI shall use Tkinter.

D2-CR-02 The numerical core shall not call built-in or library mathematical functions.

D2-CR-03 The application shall run from a terminal with `python3 beta_function_gui.py`, independent of an IDE.

D2-CR-04 Each handled input error shall name the field and the correction needed.

Quality requirements

D2-QR-01 Relative error shall be at most 10^{-10} for $B(1, 1) = 1$ and $B(2, 3) = 1/12$; $B(2, 3)$ and $B(3, 2)$ shall agree within the same tolerance.

D2-QR-02 For $0 < x, y \leq 1000$ with a representable result, the calculation shall complete within 100 ms on the demonstration computer.

D2-QR-03 After a handled error, the GUI shall remain open and ready for corrected input.

Traceability: P5 architecture satisfies D2-CR-01–04; the verification table supports D2-QR-01; GUI recovery supports D2-FR-06 and D2-QR-03. (ISO/IEC/IEEE, 2018; Kamthan, 2026a)

GAI Use: CASTROFF Evidence for P5–P7

Tool and CASTROFF

Tool: OpenAI ChatGPT/Codex.

Context: SOEN 6011 D2, F6.

Audience: TA and instructor.

Style: concise, evidence-based.

Task: implement and document P5–P7.

Role: software engineering student.

Output: Python, README, requirements, slides.

Format: runnable files and LaTeX.

Feedback: verify mathematically and by execution.

Prompt examples and decisions

P5 prompt: “Design Beta from scratch without mathematical libraries, including subordinate functions and exceptions.”

Revision: chose recurrence + Stirling, added range checks, and rejected direct Gamma multiplication.

P6 prompt: “Propose purpose-focused commits and a minimal README for this scientific GUI.”

Accepted after check: four commits match actual repository changes; README commands were executed locally.

P7 prompt: “Revise D1 shall-statements for the D2 GUI and from-scratch constraint.”

Revision: separated functional, constraint, accuracy, timing, and recovery requirements.

References

Kamthan, P. (2026a). *SOEN 6011 Project Description*. Concordia University, Summer 2026.

Kamthan, P. (2026b). *Introduction to Software Processes*. SOEN 6011 lecture notes.

Kamthan, P. (2026c). *Introduction to Software Engineering Documentation*. SOEN 6011 lecture notes.

NIST Digital Library of Mathematical Functions. (2026a). Section 5.12, Beta Function, Release 1.2.7.

NIST Digital Library of Mathematical Functions. (2026b). Section 5.2, Gamma Function Definitions, Release 1.2.7.

ISO/IEC/IEEE. (2018). *29148:2018 Systems and Software Engineering – Life Cycle Processes – Requirements Engineering*.